

YU XIAN, China

WMO: 535930

Lat: 39.8359N

Lon: 114.5723E

Elev: 910

StdP: 90.86

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-17.9	-28.0	0.3	-14.4	-26.3	0.4	-13.0	7.5	-5.1	6.4	-6.2	0.9	250	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	32.3	18.3	30.8	18.2	29.5	17.9	22.1	28.0	21.3	26.9	20.5	26.0	3.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	16.8	25.0	19.5	15.9	24.4	18.6	15.1	23.9	69.9	28.0	66.7	27.0	63.6	25.9	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.3	5.4	-23.6	35.1	2.1	1.9	-25.1	36.5	-26.3	37.6	-27.4	38.7	-29.0	40.1		
			-23.9	23.4	2.0	0.7	-25.4	23.9	-26.6	24.4	-27.7	24.8	-29.2	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-9.3	-4.4	3.3	11.1	17.2	21.8	23.8	22.0	16.8	9.7	0.8	-7.0
	DBStd	12.06	4.19	4.68	4.81	4.72	3.78	2.84	2.46	2.51	3.35	4.09	4.72	4.27
	HDD10.0	2118	598	404	214	43	2	0	0	1	54	277	526	
	HDD18.3	3901	857	637	467	222	66	6	1	3	65	267	527	784
	CDD10.0	1712	0	0	5	74	226	353	427	373	206	46	1	0
	CDD18.3	453	0	0	0	3	32	109	169	118	20	1	0	0
	CDH23.3	3860	0	0	1	72	387	965	1406	831	190	9	0	0
	CDH26.7	1202	0	0	0	16	108	322	500	217	39	1	0	0
Wind	WSAvg	1.9	1.5	1.8	2.3	2.6	2.6	2.2	1.8	1.7	1.6	1.7	1.6	1.6
Precipitation	PrecAvg	408	2	4	9	21	36	61	110	89	45	24	7	2
	PrecMax	616	10	18	39	94	75	126	198	175	88	58	20	9
	PrecMin	224	0	0	0	2	2	15	47	19	6	1	0	0
	PrecStd	99	2	4	8	17	19	31	39	40	22	15	6	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	11.9	20.9	29.0	31.9	33.7	34.7	32.5	30.4	24.7	16.1	6.2
		MCWB	-0.9	3.1	8.2	12.2	14.4	17.0	19.6	20.1	16.8	13.3	6.8	0.4
	2%	DB	2.2	8.2	17.0	25.2	29.5	31.8	32.6	30.4	27.4	21.7	13.4	3.7
		MCWB	-2.7	1.1	6.3	11.1	13.9	16.7	19.5	19.7	15.9	11.6	5.2	-1.5
	5%	DB	0.2	6.1	14.3	22.5	27.5	30.1	31.0	29.0	25.4	19.4	10.8	1.7
		MCWB	-4.0	-0.1	4.9	9.8	13.1	16.3	19.6	19.4	15.3	10.4	4.1	-2.9
	10%	DB	-1.8	3.8	11.8	20.0	25.4	28.5	29.5	27.7	23.6	17.3	8.4	0.0
		MCWB	-5.4	-1.3	3.7	8.9	12.5	16.1	19.6	19.0	14.6	9.6	3.0	-3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	4.0	8.8	14.4	17.0	20.4	23.5	22.8	19.5	14.7	7.8	0.7
		MCDB	4.1	9.8	18.4	22.8	25.9	28.0	30.4	28.6	25.5	22.0	12.3	6.1
	2%	WB	-2.4	1.8	7.0	12.7	15.9	19.4	22.4	22.0	18.1	12.8	6.3	-1.3
		MCDB	1.8	7.4	15.6	21.7	24.4	26.7	28.6	27.6	23.7	18.7	11.9	3.1
	5%	WB	-3.8	0.3	5.5	11.2	14.9	18.6	21.7	21.2	16.8	11.7	4.9	-2.5
		MCDB	0.0	5.3	13.4	19.9	23.5	25.6	27.6	26.3	22.7	17.4	10.0	1.4
	10%	WB	-5.3	-1.0	4.1	9.9	13.9	17.8	20.9	20.3	15.8	10.5	3.3	-3.7
		MCDB	-2.0	3.6	11.2	18.6	22.4	24.8	26.7	25.2	21.6	15.9	8.2	0.0
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.2	12.2	13.0	12.6	11.5	9.9	10.1	11.4	11.6	11.1	10.7
		MCDBR	13.6	15.0	16.1	17.1	16.2	14.4	12.5	12.0	14.4	14.8	14.8	13.0
		MCWBR	9.9	9.6	8.3	6.9	5.6	4.2	3.5	3.8	4.9	6.2	8.2	8.8
	5% WB	MCDBR	13.2	14.0	15.0	14.2	12.4	11.1	9.8	9.3	10.9	12.2	13.5	12.3
		MCWBR	9.6	9.0	7.9	6.2	4.9	3.9	3.6	3.6	4.5	5.6	7.7	8.5
Clear-Sky Solar Irradiance	taub	0.357	0.445	0.475	0.544	0.595	0.708	0.690	0.676	0.584	0.509	0.392	0.358	
	taud	2.077	1.873	1.837	1.751	1.667	1.514	1.590	1.594	1.716	1.818	2.051	2.119	
	Ebn at Noon	769	736	766	743	715	637	643	634	662	661	719	734	
	Edh at Noon	115	164	189	220	244	285	262	255	212	171	118	103	
All-Sky Solar Radiation	RadAvg	2.60	3.39	4.68	5.53	6.16	5.55	5.18	4.97	4.32	3.52	2.67	2.31	
	RadStd	0.16	0.22	0.38	0.35	0.46	0.44	0.29	0.33	0.28	0.28	0.32	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page